# Improving Motor Imagery Decoding through EEG Frequency Band Analysis

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***Abstract*—This study investigates the impact of specific electroencephalographic (EEG) frequency bands on decoding pedaling motor imagery in four healthy volunteers, relevant for Brain-Computer Interface (BCI) development. A frequency band ablation study was conducted on EEG data recorded during static and pedaling motor imagery. Preprocessing including a frequency filter bank, Common Spatial Patterns (CSP) for feature extraction, and Linear Discriminant Analysis (LDA) for classification were employed. The systematic exclusion of ten frequency bands (2-60 Hz) confirmed that the beta band (13-30 Hz), particularly 20-25 Hz and 30-35 Hz, is dominant for accurate pedaling motor imagery detection. Conversely, the alpha band (8-13 Hz) was more influential for relax task classification. Notably, removing the high gamma band (50-60 Hz) slightly improved overall accuracy and reduced processing time. These findings highlight the frequency-specific neural correlates of different mental states, providing insights for optimizing feature selection in EEG-based motor imagery decoding for future BCI applications.**

***Index Terms*—Brain-Computer Interface (BCI), Electroencephalography (EEG), Motor imagery (MI), Frequency filter bank, Common Spatial Patterns (CSP), Linear Discriminant Analysis (LDA).**

## I. Introduction

Electroencephalography (EEG)-based techniques are increasingly recognized as promising and versatile tools in the field of motor neurorehabilitation. These non-invasive methodologies offer a safe, cost-effective, and readily accessible means of recording neural activity directly from the scalp, thereby facilitating its seamless integration into personalized therapeutic interventions [1]. This is particularly crucial for addressing the complex underlying neurological impairments associated with a range of debilitating conditions, including neuro-degenerative diseases such as Parkinson's disease or Alzheimer's disease, and the often-significant motor deficits resulting from cerebrovascular events like stroke [2], [3]. The ability to monitor and modulate brain activity in real-time through EEG opens new avenues for targeted rehabilitation strategies [4].

Among the various paradigms employed in EEG-based neurorehabilitation, motor imagery (MI), the mental simulation of a motor act without overt physical execution, presents a particularly promising avenue for the development of Brain-Computer Interface (BCI)-based rehabilitation protocols [2], [4]. MI engages similar neural substrates to actual motor execution, making it a valuable tool for activating and potentially reorganizing motor-related brain areas in individuals with motor impairments. By harnessing the neural correlates of imagined movements, BCIs can provide a direct communication and control pathway between the brain and external devices, offering opportunities for motor recovery, functional restoration, and enhanced quality of life [5]–[7].

Despite the significant potential of EEG-based MI for neurorehabilitation, the accurate and robust decoding of motor imagery intentions from the complex and often noisy EEG signals remains a key obstacle hindering its widespread clinical translation [8]. The inherent variability of EEG signals, both within and between individuals, the low signal-to-noise ratio, and the subtle differences in neural activation patterns associated with different imagined movements pose significant challenges for developing reliable and generalizable decoding algorithms. Therefore, continued research focused on refining signal processing techniques, optimizing feature extraction methods such as the Common Spatial Patterns (CSP) algorithm, and developing robust classification algorithms like the Linear Discriminant Analysis (LDA) classifier is essential to unlock the full clinical potential of EEG-based motor imagery for neurorehabilitation applications. In this regard, identifying the most informative frequency bands is paramount for accurate motor imagery decoding and enhanced processing performance. The proposed method in this study offers an

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effective tool for precisely identifying these crucial bands by means of an ablation study.

## II. Materials and Methods

This section details the EEG recording protocol implemented in this study, including the specifications of the equipment utilized and the characteristics of the participant cohort. Furthermore, the analytical methodologies applied to the EEG data are outlined.

### A. Participants

The study group consisted of four able-bodied volunteers, including two females (identified as U01 and U03) and two males (identified as U02 and U04), with ages ranging from 29 to 49 years (29, 30, 30, and 49 years). Prior to their involvement, all participants received comprehensive verbal and written instructions regarding the study protocol and their rights. Each participant satisfied the predetermined inclusion criteria, reported no presence of diagnosed physical or neurological disorders, and provided written informed consent. This consent procedure was conducted in accordance with the ethical guidelines established and approved by the Ethics Committee of Hospital Los Madroños in Brunete, Madrid (Spain).

### B. Experimental setup

The EEG data acquisition was performed using a 32-channel g.NAUTILUS**PRO**Flexible EEG cap, in conjunction with a g.SCARABEO signal distribution system, manufactured by g.tec medical engineering GmbH. Wireless transmission of EEG signals to the registration computer was facilitated via a Wi-Fi HEADSET transmitter and a USB-connected BASE STATION receiver, also designed by g.tec. The reference electrode was attached to the right earlobe, and the ground electrode was positioned at FCZ according with the 10/10 electrode placement system [9] and adhering to manufacturer guidelines. Prior to electrode placement, the right earlobe was prepared with skin preparation gel to optimize signal quality. Furthermore, all electrodes were applied with EEG-specific conductive gel to ensure adequate signal quality and consistency, maintaining an impedance level below 20 kΩ at each electrode site. The signal acquisition had a 500Hz sampling rate.

During the EEG recordings, participants performed pedaling exercises on a clinical cycle ergometer (CycleMotus$^{TM}$ A4, Fourier Robotics) set to a constant rate of 15 revolutions per minute in lower-limb passive training mode. The experimental setup is illustrated in Figure 1.

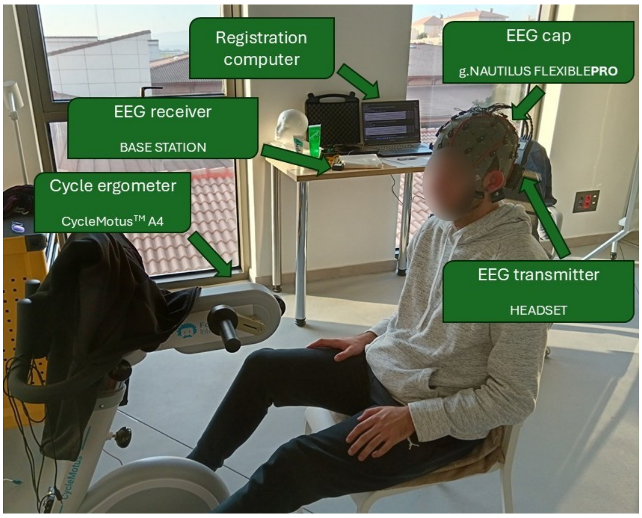


Fig. 1. Experimental setup during an EEG recording session. A participant is positioned on the cycle ergometer and fitted with the EEG cap.

### C. Experimental protocol

Each participant underwent a single EEG recording session, comprising twenty one-minute trials. Each trial was structured as follows: an initial 15-second period of mental relaxation, followed by a 30-second interval of pedaling motor imagery, and concluding with a subsequent 15-second period of relaxation. Odd-numbered trials were performed with the cycle ergometer in an inactive state, whereas even-numbered trials involved participants pedaling in passive mode. This protocol was designed to capture MI data during a static condition and relaxation data during passive cycling. The static MI data is intended to represent an activation command, while the relaxation data during pedaling corresponds to a cessation command. Furthermore, two-second epochs were introduced between each task transition to allow for the exclusion of these data segments during subsequent post-processing, thereby mitigating the potential influence of auditory cues, as participants were informed acoustically of each task change. A schematic representation of the implemented protocol is provided in Figure 2.

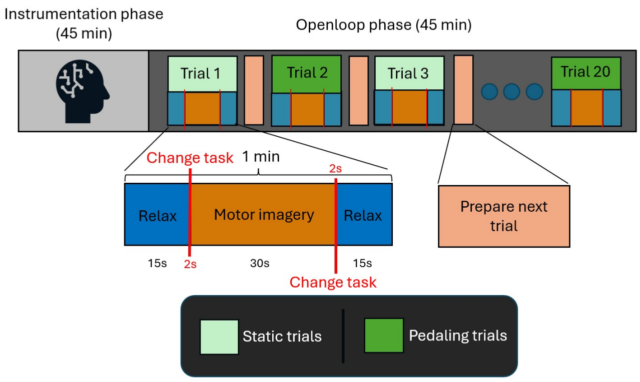


Fig. 2. This figure illustrates the experimental protocol followed in the study. Each participant completed 20 one-minute trials involving motor imagery and relaxation tasks, conducted either in a static position or while pedaling.

Prior to the commencement of each EEG recording session, all participants received detailed standardized instructions regarding the execution of the motor imagery task. These instructions emphasized the adoption of a kinesthetic approach to mentally simulate the pedaling movement, directing participants to focus on the internal muscle-sensory feedback associated with the action while actively suppressing visual imagery of the movement. This focus on kinesthetic representation was intended to elicit a more robust and localized activation of the motor cortex. Furthermore, participants were guided to achieve a clear and consistent state of mental relaxation, ensuring a homogeneous baseline condition across all trials and participants.

### *D. Motor imagery decoding algorithm*

The motor imagery decoding algorithm implemented in this study consisted of three distinct stages: preprocessing, feature extraction, and classification. All of these stages were applied after the register as an offline classification structure.

*1) Preprocessing:* The preprocessing pipeline implemented in this study involved several sequential steps. Initially, the electroencephalographic (EEG) signals underwent hardware filtering using an 8th-order band-pass filter with a frequency range of 0.5-100Hz to attenuate the influence of irrelevant frequencies [10]. Subsequently, a 48-52 Hz Notch filter was applied to mitigate artifacts arising from the 50Hz power line interference. Additionally, a noise reduction algorithm, provided by g.tec, was employed to enhance signal quality. Following these steps, a channel selection procedure was performed to isolate electrodes overlying the motor cortex, which is primarily related to motor imagery [11]. This selection resulted in 19 pre-selected EEG channels, according to the 10/10 electrode placement system [9]: FZ, F4, F3, FC5, FC1, FC2, FC6, T7, C3, Cz, C4, T8, CP5, CP1, CP2, CP6, P3, PZ, and P4. A visual representation of this electrode pre-selection is provided in Figure 3. Reducing the number of electrodes in the analysis can significantly decrease computational processing times, thereby minimizing latency, a critical factor for online applications. However, this reduction inherently leads to a loss of information, which may degrade the algorithm's classification performance. Therefore, a careful balance must be struck to optimize computational efficiency without unduly compromising algorithmic accuracy.

Following channel selection, the EEG signals were processed using a software-based frequency filter bank. This bank comprised ten distinct 2nd-order Butterworth band-pass filters, covering the following frequency ranges: 2–5Hz, 5–10Hz, 10–15Hz, 15–20Hz, 20–25Hz, 25–35Hz, 35–40Hz, 40–45Hz, 45–50Hz, and 50–60Hz. The signals from each selected channel were filtered in parallel by each of these band-pass filters, and the resulting filtered outputs were subsequently summed element-wise, adhering to the methodology described by Wang et al. [12]. A schematic representation of this filter bank is provided in Figure 4. To finalize the preprocessing stage, the filtered channels underwent a normalization procedure aimed at minimizing the impact of artifacts such as blinks, evoked potentials, and signal interferences. The mathematical definition of this normalization is presented in Equation (1).

$$EV[t] = \frac{V[t] - \overline{V}}{STD_V} \tag{1}$$

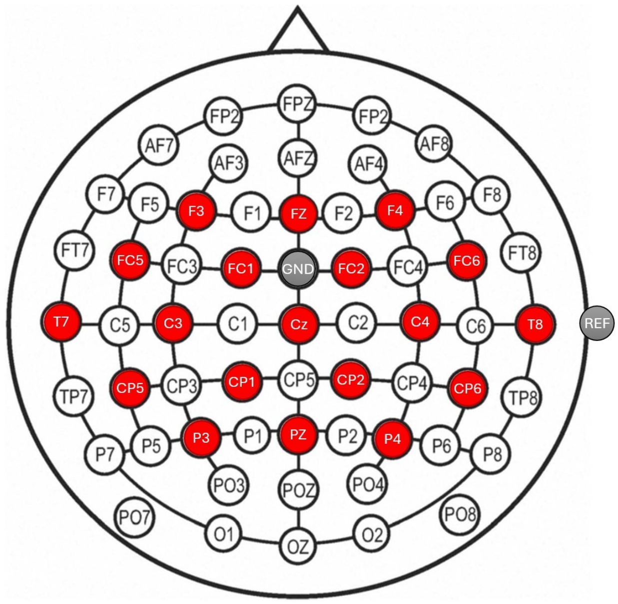


Fig. 3. Spatial distribution of the pre-selected electrodes over the motor cortex is represented in this figure. Electrodes chosen for analysis are highlighted in **red**, with the ground (GND, FCZ) and reference (REF, A2) electrodes marked in **gray**.

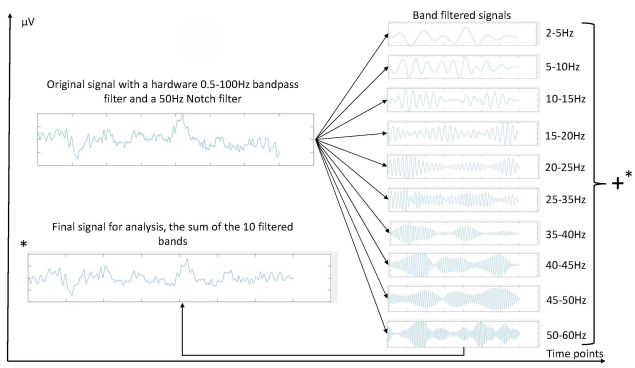


Fig. 4. This figure illustrates the application of the filter bank. The original EEG signal underwent parallel filtering across multiple frequency bands, with the resulting filtered signals subsequently summed element-wise to yield the final filtered signal.

Where $EV[t]$ denotes the normalized EEG sample for a given channel at time instant $t$, $V[t]$ represents the filtered EEG sample for that channel at time instant $t$ before normalization, $\overline{V}$ corresponds to the mean value of that channel's filtered signal across the trial, and $STD_V$ signifies the standard deviation of that channel's filtered signal over the same epoch.

*2) Feature extraction:* Upon completion of the normalization procedure, the preprocessing stage was finalized. To initiate feature extraction, the normalized signals underwent temporal segmentation. This process involved the extraction of one-second epochs with a 0.5-second overlap, a design that enabled feature computation based on the most recent one-second segment of the signal every half-second, thereby yielding classification outputs at 0.5-second intervals. Any window containing task transition data, either partially or entirely, was excluded from the analysis. This temporal segmentation was applied to all selected channels.

Feature extraction in this study employed spectro-spatial features derived through the Common Spatial Patterns (CSP) algorithm. CSP is a well-established multivariate signal processing technique specifically designed to identify spatial filters that maximize the separability between distinct classes within multichannel data, such as electroencephalographic recordings. This is achieved by computing spatial filters that project the original sensor data into a new subspace where the variance of the signal differs maximally between the classes of interest, effectively enhancing class discriminability. In the context of our temporally segmented EEG data, the CSP algorithm was applied to each one-second epoch extracted. For each epoch and each participant, CSP filters were trained to discriminate between the two mental conditions (relax and MI) based on the normalized multi-channel EEG signals within that specific time window. This epoch-wise application of CSP allowed for the extraction of dynamic spectro-spatial features that capture the temporal evolution of the brain activity patterns associated with the different motor imagery tasks.

*3) Classification:* Following the temporal segmentation of the preprocessed EEG signals and the subsequent feature extraction, the classification stage was implemented. This component of the MI decoding algorithm employed a Linear Discriminant Analysis (LDA) classifier. Linear Discriminant Analysis is a supervised machine learning algorithm commonly utilized for classification tasks. It functions by identifying a linear combination of features that optimally discriminates between two or more distinct classes. This is achieved by projecting the data onto a lower-dimensional subspace in a manner that maximizes the inter-class variance while minimizing the intra-class variance. The LDA classifier was trained using the spectro-spatial features generated in the preceding feature extraction step. Notably, the classification models adopted an intra-subject architecture; that is, separate models were trained for each individual participant, thereby avoiding the incorporation of data from other participants due to the substantial inter-subject variability inherent in EEG signals.

Furthermore, distinct classification models were developed for the two experimental conditions: static and cycling. For the static models, the Linear Discriminant Analysis (LDA) classifier was exclusively trained using the spectro-spatial features extracted from the static trials. This process was repeated to create separate cycling models, where the classifier was trained solely on features derived from the trials performed during passive pedaling. This segregated modeling approach aims to generate more specialized and consequently more efficient decoding models for potential real-time applications. In such scenarios, the state of the cycle ergometer (active or inactive) would be readily available, enabling the selection and application of the most appropriate decoding model at any given time. To obtain robust performance estimates and derive more reliable conclusions from this study, a leave-one-out cross-fold validation procedure was also implemented for each individual model and participant. Given that each participant completed ten static motor imagery trials and ten cycling motor imagery trials, the classification process was repeated ten times for every model (static and cycling) and for each user. In each iteration, a single trial was withheld as the test set, while the remaining nine trials were used for training the LDA classifier. This iterative validation technique provides a more reliable assessment of the model's generalization capability by evaluating its performance on each individual trial as an independent test case. There is a representation of this validation process in the Figure 5.

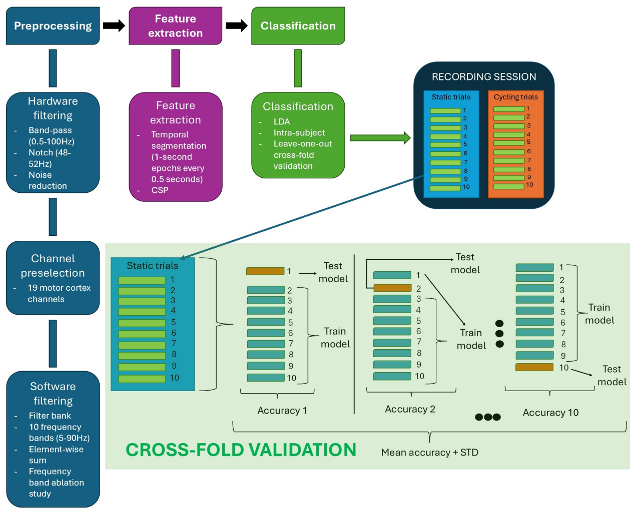


Fig. 5. This figure presents a schematic overview of the entire EEG processing pipeline implemented in this study. The motor imagery decoding algorithm comprises a preprocessing stage, a feature extraction process utilizing CSP, and classification via LDA. A cross-validation framework was also applied to obtain standard deviation (STD) metrics for performance evaluation. Furthermore, this work includes a frequency band ablation study.

## *E. Frequency band ablation study*

Following the computation of the classification results detailed in the preceding subsection, the entire motor imagery decoding algorithm (excluding the initial hardware filtering stage) was reiterated ten times for each classification model. In each iteration, one of the ten frequency bands comprising the filter bank in the preprocessing step was systematically excluded. This procedure constituted a frequency band ablation study, enabling the quantitative assessment of each frequency band's contribution to classification performance by comparing the resulting metrics with those obtained from the baseline model (utilizing all frequency bands). A comprehensive schematic of the complete motor imagery decoding algorithm, including a graphical representation of the cross-fold validation process, is presented in Figure 5.

## III. Results

Table I presents a comprehensive summary of the classification accuracies achieved in this study, including the baseline performance where no frequency band was omitted from the filter bank. The values reported in the table represent the average classification accuracies across the four participants. Overall, the average classification accuracy across all conditions exceeded 70%, indicating a generally balanced performance between the motor imagery and relaxation classes. However, a notable degree of variability was observed in the classification accuracy across individual trials, with the average standard deviation generated in the cross-fold validation process surpassing 17%. The results are also presented in Figure 6 for further clarity.

TABLE I
COMPARISON OF THE CSP ACCURACIES AND STANDARD DEVIATION (STD) RESULTS DEPENDING ON THE OMITTED FREQUENCY BAND REGARDING THE BASELINE THAT INCLUDES ALL THE BANDS

| Omitted frequency band | RELAX Accuracy [%] | MI Accuracy [%] | GLOBAL STD |
|---|---|---|---|
| 5-10Hz | 70.7 | 75.3 | 19.7 |
| 10-15Hz | 63.4 | 70.8 | 14.5 |
| 15-20Hz | 68.3 | 67.6 | 15.9 |
| 20-25Hz | 70.7 | 65.0 | 14.3 |
| 30-35Hz | 70.6 | 62.9 | 14.1 |
| 35-45Hz | 80.0 | 67.3 | 17.5 |
| 55-60Hz | 69.0 | 69.1 | 15.5 |
| 60-70Hz | 72.8 | 74.9 | 19.5 |
| 70-80Hz | 69.6 | 70.6 | 19.5 |
| 80-90Hz | 73.8 | 74.1 | 20.2 |
| **Baseline** | **71.3** | **70.4** | **21.2** |

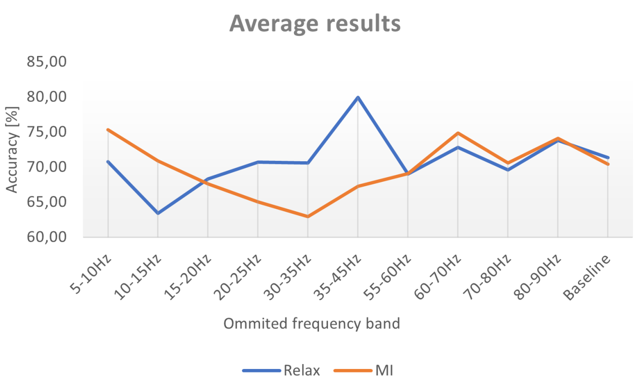


Fig. 6. Average results of the study are presented in this figure. The **blue line** represents the average relax classification accuracy observed upon the exclusion of each individual frequency band across subjects. Similarly, the **orange line** depicts the average MI classification accuracy. The baseline that includes all the bands is also presented.

The results of the frequency bands ablation study revealed distinct impacts of specific frequency bands on the classification of the two mental states, and they are summarized in Table II. Excluding the alpha frequency band (approximately 8–13 Hz) led to a discernible decrease in the average classification accuracy for the relaxation task, with a reduction of nearly 8% observed within the 10–15 Hz range.

TABLE II
DIFFERENCES OF THE CSP ACCURACIES AND STANDARD DEVIATION (STD) RESULTS DEPENDING ON THE OMITTED FREQUENCY BAND WITH THE BASELINE THAT INCLUDES ALL THE BANDS

| Omitted frequency band | RELAX Accuracy [%] | MI Accuracy [%] | GLOBAL STD |
|---|---|---|---|
| 5-10Hz | -0.6 | 4.9 | -1.6 |
| 10-15Hz | -8.0 | 0.4 | -6.8 |
| 15-20Hz | -3.0 | -2.8 | -5.3 |
| 20-25Hz | -0.7 | -5.4 | -6.9 |
| 30-35Hz | -0.8 | -7.5 | -7.1 |
| 35-45Hz | 8.6 | -3.2 | -3.7 |
| 55-60Hz | -2.3 | -1.4 | -5.6 |
| 60-70Hz | 1.5 | 4.4 | -1.7 |
| 70-80Hz | -1.7 | 0.2 | -1.7 |
| 80-90Hz | 2.4 | 3.7 | -1.0 |

Conversely, the classification accuracy for the motor imagery task was more significantly affected by the exclusion of the beta frequency band (typically around 13–30 Hz). The absence of the 20–25 Hz band resulted in an average MI classification accuracy reduction exceeding 5%, while the exclusion of the 30–35 Hz band led to an even more substantial decrease of over 7.5%. Figure 7 depicts all these changes with the baseline.

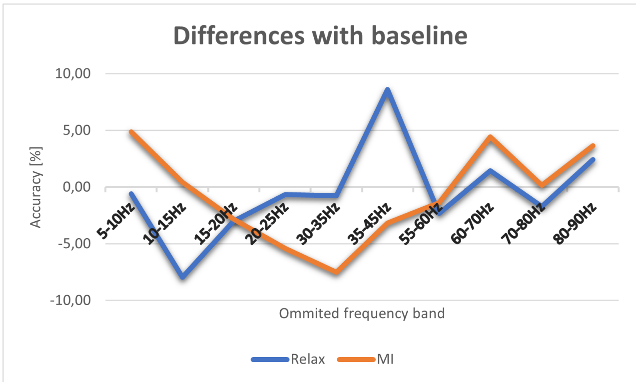


Fig. 7. Results of the ablation study are presented in this figure. The **blue line** represents the changes in relax classification accuracy observed upon the exclusion of each individual frequency band, relative to the baseline accuracy obtained using the complete filter bank. Similarly, the **orange line** depicts the corresponding changes in MI classification accuracy.

Interestingly, the ablation of the highest frequency band (50–60 Hz), which is generally considered to carry less relevant information for motor imagery EEG signals, resulted in an average improvement in overall classification accuracy of over 3%. Furthermore, this omission was associated with a reduction in computational processing times.

Finally, the ablation of any single frequency band consistently led to a reduction in the inter-trial and inter-subject variability of the classification accuracy. The most pronounced reductions in standard deviation (exceeding 7%) were observed when either the alpha or the beta frequency bands were removed from the filter bank.

## IV. Conclusions and discussion

The average classification accuracies exceeding 70% demonstrate the feasibility of decoding static and pedaling motor imagery from EEG signals using the implemented methodology. The relatively balanced performance between the two classes suggests that the extracted spectro-spatial features and the Linear Discriminant Analysis classifier were reasonably effective in distinguishing these mental states. However, the significant inter-trial variability (average standard deviation over 17%) indicates a potential area for further investigation and refinement of the decoding algorithm or the experimental protocol to enhance robustness and consistency.

The observed decrease in relaxation classification accuracy upon exclusion of the alpha band aligns with existing literature [10] suggesting the involvement of alpha oscillations in attentional and relaxing brain states, which are likely modulated during mental relaxation. Similarly, the decline in motor imagery classification accuracy when the beta band was excluded corroborates previous findings [10] highlighting the crucial role of beta rhythms in motor planning and execution, even during imagined movements.

The notable improvement in average classification accuracy and the reduction in processing time upon the exclusion of the highest frequency band (50–60 Hz) suggest that this band may primarily contribute noise or irrelevant information to the decoding process in this specific context. This finding has practical implications for real-time BCI applications, as removing this band could lead to more efficient and potentially more accurate systems.

The consistent reduction in inter-trial and inter-subject variability when individual frequency bands were omitted, particularly alpha and beta, warrants further consideration. The initial hypothesis that this might be due to a lack of determinant features leading to more biased results could be explored in more detail. It is possible that the presence of these key frequency bands contributes to a wider range of neural responses across trials, leading to higher variability. Conversely, their absence might force the classifier to rely on less specific features, resulting in more consistent but potentially less accurate outcomes in some cases. Future research could investigate the specific neural dynamics within these frequency bands during motor imagery and relaxation to better understand their contribution to classification variability.

Future work should focus on obtaining a higher degree of statistical evidence, which necessitates including a larger cohort of participants in the study. This expanded research should specifically investigate the frequency bands of interest, primarily 20-25 Hz, 30-35 Hz, and 50-60 Hz. Further investigation into the average classification improvement achieved by omitting the 50-60 Hz band could clarify key aspects in motor imagery decoding and significantly enhance the performance of future decoding models.

In conclusion, this study provides valuable insights into the decoding of static and pedaling motor imagery using EEG. The findings regarding the importance of alpha and beta bands, as well as the potential benefits of excluding high-frequency noise, contribute to the growing body of knowledge in motor imagery-based BCIs and offer potential avenues for optimizing future real-time applications. The proposed methodology has led to an increase in decoding accuracy and a decrease in processing times, thereby demonstrating its effectiveness, particularly for online classification applications. Consequently, it holds significant potential for employment in neurorehabilitation therapies.

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